W-C-20E
July 29, 1980
SUPERSEDING
Fed. Spec. W-C-20D
May 20, 1970

FEDERAL SPECIFICATION

CABINET, DOUGH-PROOFING, (18 PAN CAPACITY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a mobile dough-proofing cabinet of one type and size with an electrically heated humidifier.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

FF-C-77 - Casters, Rigid and Swivel (Institutional Duty)

Federal Standard:

FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative biomonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Washington, DC; Philadelphia; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

FSC 7320

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Military Specification:

MIL-B-2316 - Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery of

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Sanitation Foundation (NSF) Standard

No. 4 - Gas and Electric Commercial Cooking and Warming Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

Underwriters Laboratories Inc. (UL) Standard

UL 197 - Commercial Electric Cooking Appliance

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Standard product. The proofing cabinet shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incoroporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturers or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The proofing cabinet shall comply with the applicable requirements of NSF Standard No. 4 and UL Standard No. 197.

3.2.1 Compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the proofing cabinets he proposes to furnish under this specification meet the applicable requirements of NSF and UL, as applicable.

3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

(1) A listing in the currect edition of the NSF "Listing of Food Service Equipment" and a display of the NSF seal on the finished proofing cabinet.

(2) A certification for the proofing cabinet issued by NSF under their one-time contract evaluation/certification service shall be submitted.

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical authority functioning in an advisory capacity to the preparing agency of the specification, indicating that the proofing cabinet has been tested and conforms to NSF Standard No. 4.

3.2.1.2 UL. Acceptable evidence of meeting the requirements of UL 197 for the proofing cabinet shall be the UL Listing Mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating compliance with the applicable requirements of UL 197.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial proofing cabinets provided the complete item complies with all provisions of this specification (see 6.3).

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3.4 Design and construction. The dough-proofing cabinet shall be fabricated of aluminum alloy. The cabinet shall accommodate a minimum of 18 pastry pans having nominal overall dimensions of 18 inches (457 mm) by 26 inches (660 mm) by 1-1/4 inches (32 mm) deep. The cabinet maximum overall dimensions, including casters, shall be 78 inches (2 mm) high, 24-1/2 inches (622 mm) wide, and 34 inches (864 mm) deep. The ledges or guides which support the pans shall be spaced 3 inches (76 mm), plus or minus 1/8 inch (3 mm), on centers supporting pans at the rims or bottoms to prevent droppage and spillage of the loaded pans. The cabinet shall be equipped with an electrically heated humidifier (see 3.4.3). The cabinet shall have spaces to allow steam to circulate throughout the cabinet interior with the pans in place. The cabinet shall be provided with four removable casters as specified in 3.4.5 with two casters equipped with foot brakes. The casters with foot brakes shall be located at the front of the cabinet. The cabinet ledges, and caster mounting areas shall not show any evidence of damage or failure when tested as specified in 4.4.1.

3.4.1 Frame and enclosure. The cabinet shall be of a frame or frameless construction. When a frame is not used the back and sides shall be formed from a single sheet of aluminum to form the enclosure. Top and bottom sections of the frame less construction shall be either welded or riveted to the sides. When frame construction is used, the cabinet shall be constructed of a frame and panels. The panels shall be riveted or welded to the frame to form the sides, back, top and bottom of the cabinets. The bottom of cabinet shall be reinforced for caster mounting.

3.4.2 Door. A full length door shall provide access to the pans. The door shall be of the same material as the enclosure and shall be hinged to permit it to be opened a minimum of 180 degrees. A positive acting latch shall be provided. The door shall have a minimum of three separate hinges. Hinges and attaching hardware shall be stainless steel or aluminum alloy. The doors shall close tightly against the doorjamb or cabinet frame. The door edges shall be reinforced by structural shapes or forming of the panels. Provisions shall be made on the cabinet to hold the door in an open position. A thermometer shall be mounted on the outside of the door to indicate the air temperature inside the proofing cabinet without opening the door. The door shall not deform when tested as specified in 4.4.2.

3.4.3 Humidifier. A removable humidifier shall be located in the bottom of the cabinet. The humidifier shall be fabricated of corrosion-resistant metal. The water pan shall be constructed of stainless steel. A cover with a locking device shall be provided for the water pan to prevent water from splashing when the cabinet is moved. The water pan shall have a minimum capacity of 3-1/2 quarts (3.3 L) and shall be heated by an immersion type heater. Unless otherwise specified (see 6.2), the cabinet including heater, shall be designed for use on a nominal 120-volt, 50/60 Hertz 1-phase system. An automatic thermostat shall be provided to control the temperature of the water. A handset (manual) mechanical timer, with an "ON-OFF" switch having a time limit of at least 10 hours

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shall be provided. A pilot light shall be provided to indicate when electrical power is on. The humidifier shall be capable of maintaining 85 to 100 percent relative humidity with a cabinet temperature of 90° to 100°F (32.2° to 37.7°C). Heat distribution shall not vary more than plus or minus 5°F (3°C) when tested in accordance with 4.4.3.

3.4.4 Handles. Two drop-type trunk handles shall be provided on each side of the cabinet, mounted at a convenient height for moving and lifting the cabinet. The handle brackets shall be designed to limit the swing of the handle to 90° from the normal position of being folded flush against the cabinet wall. The bail of the handle shall be not less than 4 inches (102 mm) wide (inside), and shall be fabricated from not less than 3/8 inch (10 mm) diameter rod, either stainless steel or aluminum alloy.

3.4.5 Casters. Casters shall conform to FF-C-77, type II, class B, wheel diameter 5 inches (127 mm), style c with a minimum tread width of 1-1/4 inches (32 mm), and mounting P. The casters shall not deform when tested as specified in 4.4.2.

3.4.6 Bumpers. The proofing cabinet shall be equipped with corner bumpers or continuous wrap-around (perimeter) bumpers. Bumpers shall be fabricated from rubber or rubber-like material and be assembled in accordance with the manufacturers standard practice. No part of the proofing cabinet except the bumper(s) shall touch a verticle wall upon contact.

3.5 Welding. When applicable, all surfaces to be welded shall be free from oxide, scale, and grease. Welds shall be continuous, sound and smooth and free from incomplete fusion, porosity, cracks, undercuts, burn-through and deformation of material. All flux deposits (when flux is used) shall be removed from welds.

3.6 Marking. Each cabinet shall be permanently and legibly marked with the manufacturer's model and serial number, name, trade name or trademark of such known character as to be identifiable with said manufacturer.

3.7 Finish. The cabinet shall be smooth on all surfaces and edges; free from discoloration or stains and shall be finished in accordance with the manufacturers standard practice.

3.8 Workmanship. All kinks, bends, and twists shall be removed from sheet metal used in the fabrication of the cabinet. The cabinet shall not contain any rough or sharp edges, burrs, or slivers, and shall be clean and free from rough grind, die, and trimming marks, rust, and scale. Welding shall be accomplished in a manner that will prevent the occurrence of pits or flashes. The finished item shall not be fractured, buckled, dented, punctured, or malformed. Riveting shall be accomplished so that when the peening or upsetting operation is performed the driven rivet shall completely fill the hole and the head and upset end shall make full contact with the surface members. When bolts, nuts, or threaded fasteners are used they shall be drawn up tight.

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4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except where otherwise indicated herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall consist of all proofing cabinets offered for inspection at one time. The sample unit shall be one completely fabricated proofing cabinet.

4.2.2.1 Visual examination. Examination of the proofing cabinets shall be made for the defects in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Finish discolored or stained.	X	
Design, construction and workmanship	Cabinet not of welded or rivited frame construction.	X	
	Any kinks, unspecified bends, or twist on metals.	X	
	Any burr, sliver, sharp edge, or any rough die, tool, gouge, trim or grind marks.	X	
	Any component fractured, buckled, dented, punctured, or malformed.	X	
	No spaces to allow circulation of steam in the cabinet interior when the pans are in place.	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Frame and enclosure	Sides and back not formed to form the enclosure (frameless construction).	X	
	Ledges or guides do not support pans by rims or bottom.	X	
	Top and bottom sections not welded or riveted to sides and frames.	X	
	Bottom exterior corners not reinforced for casters.	X	
Door	Not as specified.	X	
	Door edge not reinforced.	X	
	Not a minimum of three separate hinges.	X	
	No provisions on walls of cabinet for holding door in open position.	X	
	Door does not close tightly against door jam or cabinet frame.	X	
	Not provided with a positive acting latch.	X	
	Not provided with thermometer.	X	
Humidifier	Not as specified.	X	
	Not movable.	X	
	Cover on water pan not provided with locking device.	X	
	Water pan cover locking device missing.	X	
Timer	Not "ON-OFF" type.	X	
	Not manual handset type.	X	
	Does not have minimum time limit of 10 hours.		X
	Pilot light missing.	X	
Handles	Not trunk type.	X	
	Trunk type handle does not fold flush against wall of cabinet	X	
	Not located on each side of cabinet.	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Casters	Not swivel type.	X	
	Foot actuated brakes missing; on adjacent casters.	X	
	Not removable.	X	
	Not attached as specified.	X	
Bumpers	Missing or not as specified.	X	
Welding (when applicable)	Not free from oxide, scale, or grease.		X
	Excessive flux not removed.		X
	Not continuous, sound, smooth; or free from pits, flash, porosity, cracks, incomplete fusion, undercut, or deformation of material.	X	
Riveting (when applicable)	Rivet, when upset, does not make full contact with surface members.		X
	Rivet not neatly set or secure that is loose or removable.	X	
Bolts, nuts, screws, studs and other threaded fasteners	Missing, broken, stripped, or fractured.		X
	Not drawn up tight.		X
Identification marking	Missing, illegible, incomplete; or not permanently attached.	X	

4.2.2.2 Dimensional examination. The proofing cabinets shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in defects per hundred units.

4.2.2.3 End item testing. Each proofing cabinet shall be tested as specified in 4.4.1. Failure to pass the test shall be cause for rejection of the proofing cabinet. The proofing cabinets shall be tested as specified in 4.4.2 and 4.4.3, using an inspection level of S-2 with an AQL of 4.0 defects, expressed in defects per hundred units.

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4.2.3 Examination of preparation for delivery. Examination shall be made for preservation, packaging, packing and marking in accordance with MIL-B-2316.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Test methods.

4.4.1 Obstacle test. The cabinet shall be filled to capacity with bun pans. A minimum load of 250 pounds (113 kg) shall be evenly distributed throughout the pans. Each pan shall have the weight equally distributed over its entire surface. The loaded cabinet shall pass over a 1/4 inch (6 mm) obstacle ten times at a minimum speed of 2 miles (3.2 km) per hour to determine compliance with 3.5. Any nonconformance shall constitute failure of this test.

4.4.2 Deformation test. After the test in 4.4.1 has been accomplished the loaded cabinet shall stand in one position for 8 hours. When examined at the end of 8 hours, the door and casters shall be examined for compliance with 3.4.2 and 3.4.5, respectively. Any nonconformance shall constitute failure of this test.

4.4.3 Heat distribution and humidity test. Thermocouples shall be located at three equidistant vertical points in the cabinet. The cabinet shall be connected to a source of power having the same electrical characteristics as specified in the contract order. With the door closed, the humidifier shall operate for one hour. After one hour of operation temperature shall be read on a recording potentiometer and the humidity read on a humidity recording instrument to determine compliance with 3.4.3. Any nonconformance shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The cabinets shall be preserved and packaged in accordance with the level A packaging requirements of MIL-B-2316.

5.1.2 Commercial. The cabinets shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the cabinet against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

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5.2.1 Level A packing. The cabinets, packaged as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-B-2316.

5.2.2 Level B packing. The cabinets, packaged as specified in 5.1, shall be in accordance with the level B packing requirements of MIL-B-2316.

5.2.3 Commercial packing. The cabinets, packaged as specified in 5.1, shall be packed to insure delivery at destination, provide for redistribution by the initial receiving activity and be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Marking shall be in accordance with 5.2.1 or 5.2.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military activities. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dough-proofing cabinet is intended for use in bakeries, central pastry kitchens, and dining facilities kitchens for pannery proofing of yeast raised molded dough pieces prior to being loaded into the oven for baking.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Electrical requirements of humidifier, if other than specified (see 3.4.3).
(c) Selection of applicable levels of preservation, packaging, packing, and marking (see 5.1, 5.2, and 5.3).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

6.4 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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